IVRHA SYMPOSIUM · FROM MICROGRAVITY TO MEDICINE

Designing a Virtual-Reality Neuro-Visual Assessment Platform for Spaceflight-Associated Neuro-Ocular Syndrome

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Outline

01

The problem

Why SANS threatens astronaut vision, and why eye exams in flight are hard

02

The platform

One VR headset that runs five standardized eye tests

03

Going to space

What happened when microgravity broke the headset's tracking

04

Lessons learned

Four design lessons for VR tools in altered gravity

01

The problem

SANS threatens astronaut vision, and checking eyes in flight is hard

Long duration space-flights (LDSF) change astronauts' eyes

48%

of astronauts had worse near vision after long flights, vs 23% after short ones[1]

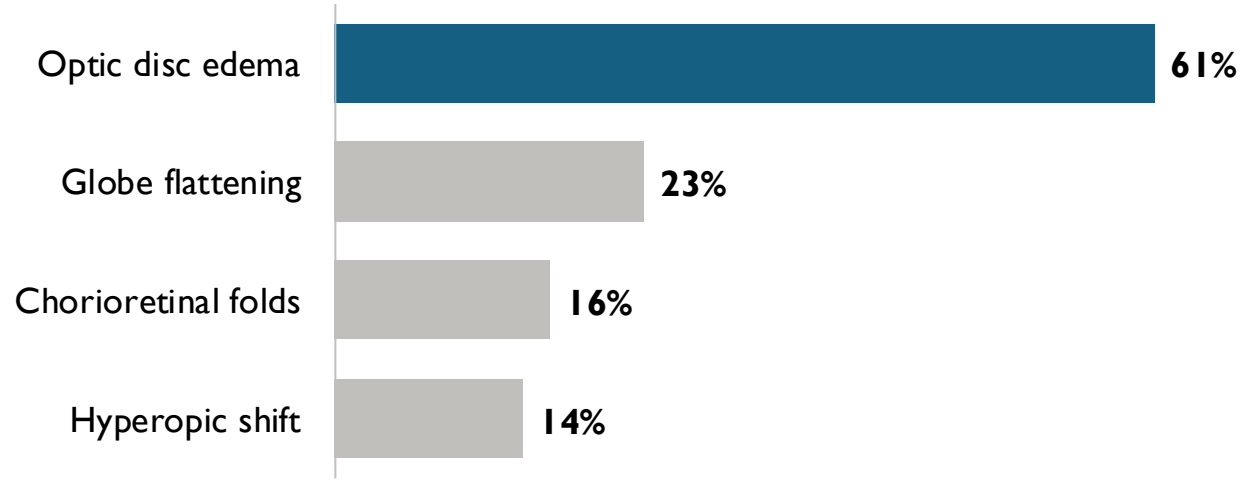
- **Changes appear early** in flight and can persist for years[1]
- **NASA rates these eye changes a top risk** for a crewed mission to Mars[2]
- **Mars and moon crews will spend longer** in weightlessness than anyone has before[2]

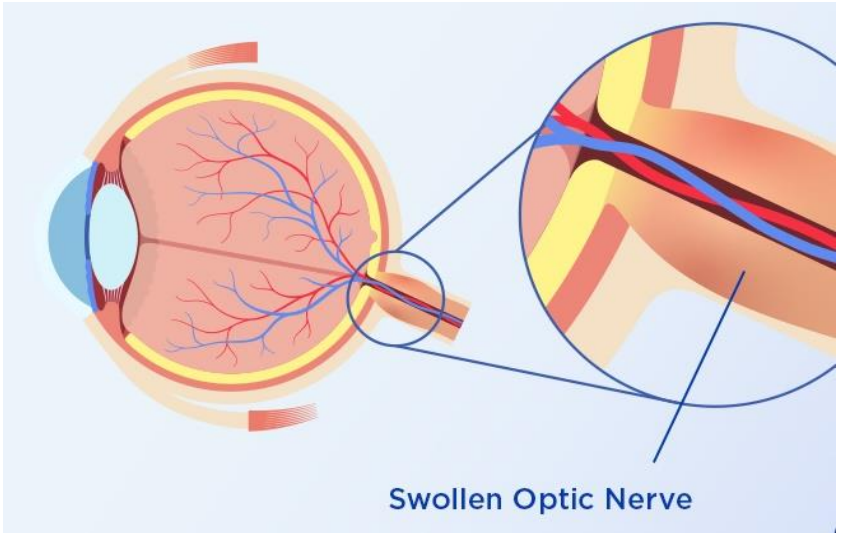
1. Nguyen et al. (2025). 2. Patel et al. (2020).

Spaceflight-Associated Neuro-ocular Syndrome (SANS)

Constellation of eye and optic nerve changes in astronauts on missions of 30 days or more[1]

Astronauts with each SANS sign (NASA earliest-indication threshold)

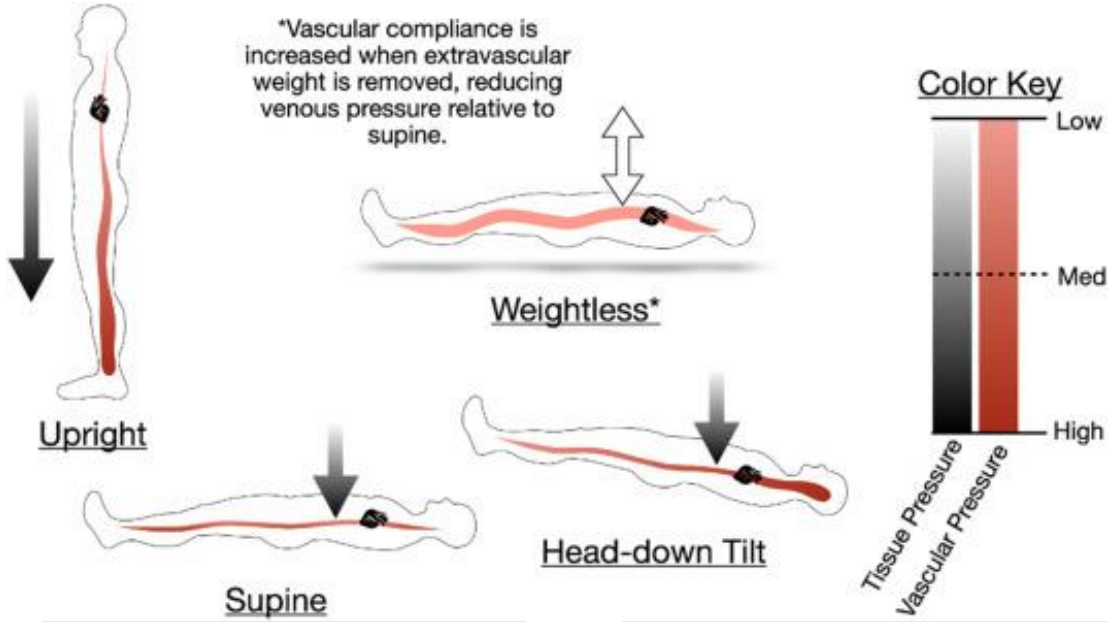




Papilledema, swollen optic nerve

1. Nguyen et al. (2025), Table 1.
Image: https://www.oscarwylee.com.au/glasses/eye/papilledema

The leading explanation is a headward fluid shift



01

Microgravity

removes the head-to-foot pressure gradient

→

02

Headward shift

blood and CSF move toward the head

→

03

↑ Intracranial Pressure (ICP)

pressure builds around the brain and optic nerve

→

04

Eye remodels

disc swells, globe flattens, folds form

Sources: Nguyen et al. (2025), Section 6; Waisberg et al. (2023). Mechanisms are proposed and not mutually exclusive.

Previous in-flight eye checks rely on high-contrast tests



Routine checks on the station

Laptop static visual acuity, a self-reported survey and an Amsler grid



May miss optic nerve disease

These high-contrast tests may not be sensitive enough to catch optic nerve disease



Contrast only when indicated

Contrast sensitivity can be tested on board, but only when clinically indicated

Source: Waisberg et al. (2023), Front. Ophthalmol.

02

The platform

One headset that standardizes five eye tests and travels light

Reasons for using VR

STANDARDIZATION

Every test gets the same light and measurements

Typical exams are conducted in differently lit rooms.

Subjective, rather than objective measurements

EXAMINER

The astronaut can run it alone

No trained examiner is needed: the exam is self-operated.

EQUIPMENT

Many tests in one headset

Clinic devices are stationary and usually test one thing. One portable headset runs all five tests.

Source: Waisberg et al. (2023)

Our platform: one headset, five tests, both eyes tracked

THE PLATFORM

Headset
Pico 4E

Eye tracking
Infrared VGA, OV6211, 400 × 400 @60Hz/90Hz, H70°/V70°

Software
Built in Unity 6 on our ProjektPerception SDK

Tests
Acuity, contrast, color, pupil reflex (RAPD), and visual field

> **Program developed at UNR** in collaboration with NASA



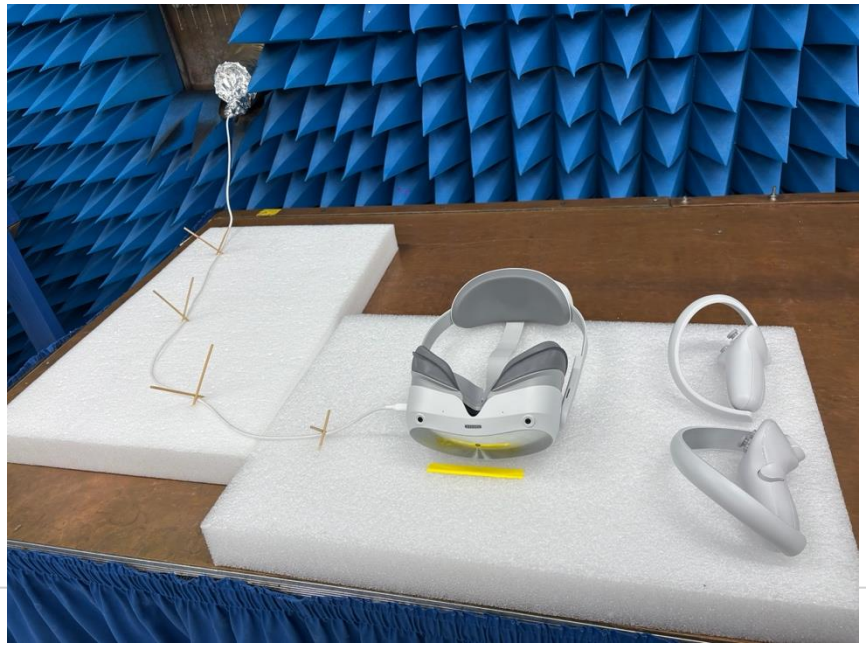
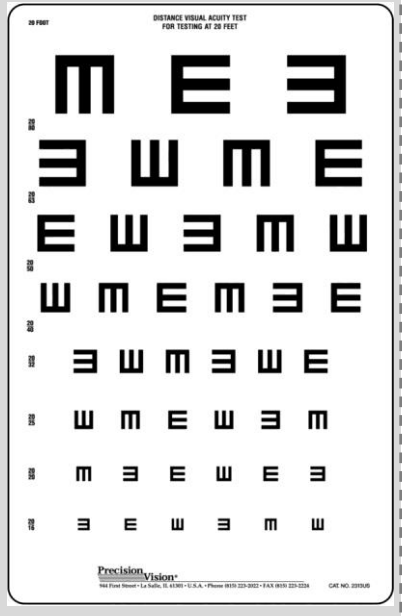


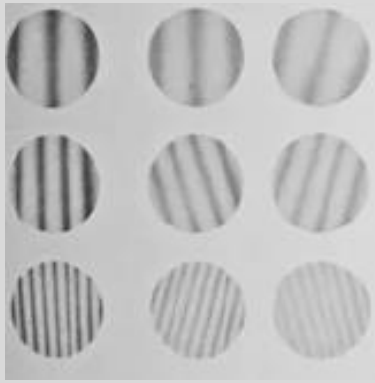
Image Source: Amazon.com (Top); Personal (Bottom)

What the tests look like inside the headset

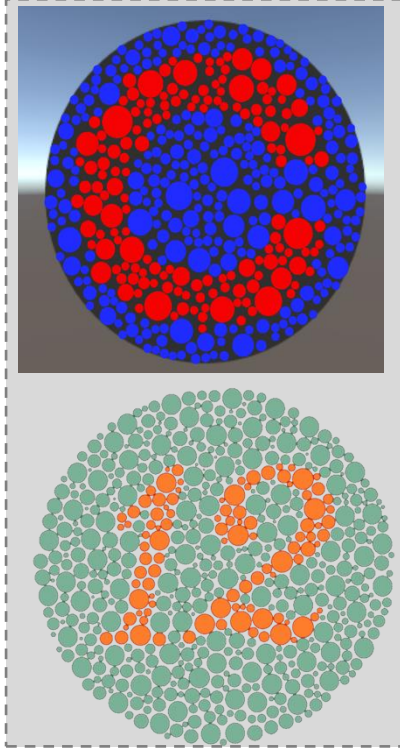
ACUITY



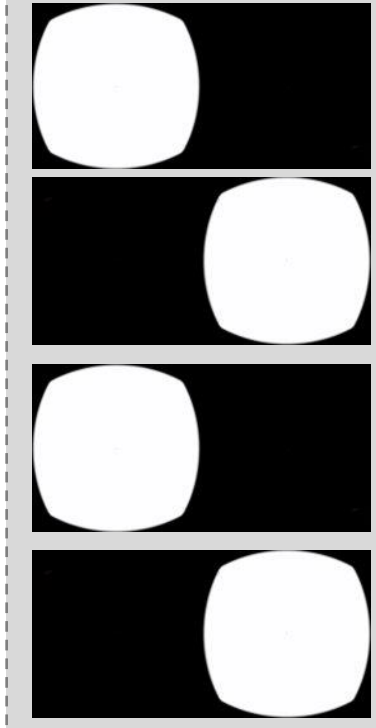
CONTRAST



COLOR



RAPD



VISUAL FIELD



Image Source: Acuity - https://eyewiki.org/Visual_Acuity_Assessment_in_Children; Other Images: myself

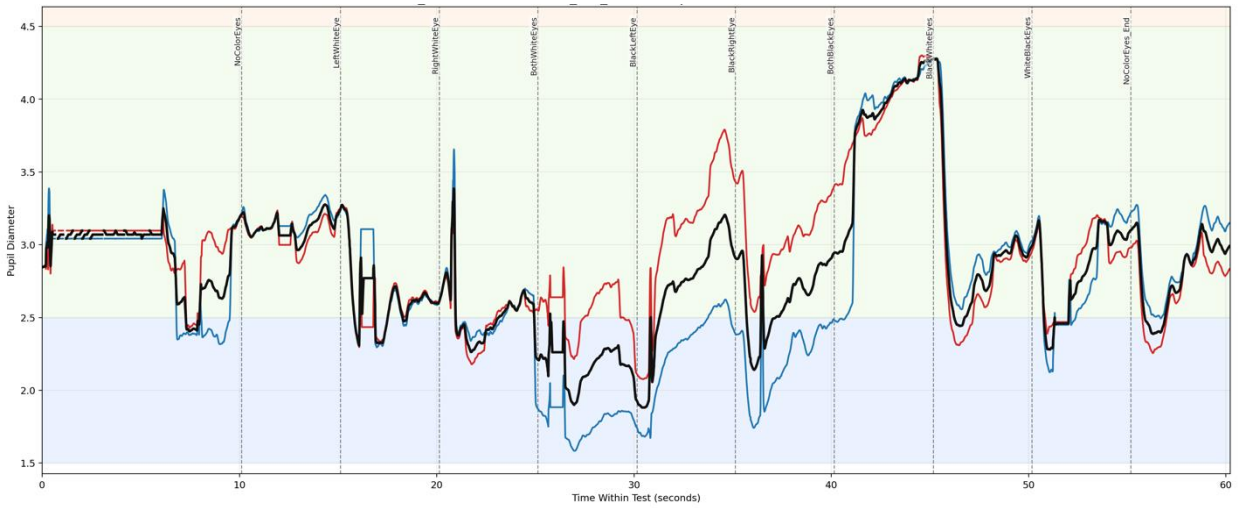
Each test is built to detect a specific SANS change

OUR TEST	SANS CHANGE IT IS MEANT TO DETECT
Visual acuity (tumbling E)	Globe flattening → farsighted shift
Contrast sensitivity (sinusoidal gratings)	Possible retinal ganglion cell damage from prolonged disc swelling
Color vision (Cambridge Color Test; Ishihara screening)	Optic nerve function
RAPD / pupillometry (swinging flashlight)	Uneven disc swelling; pupil–ICP link
Visual field perimetry (24-2)	Disc swelling; peripheral field loss

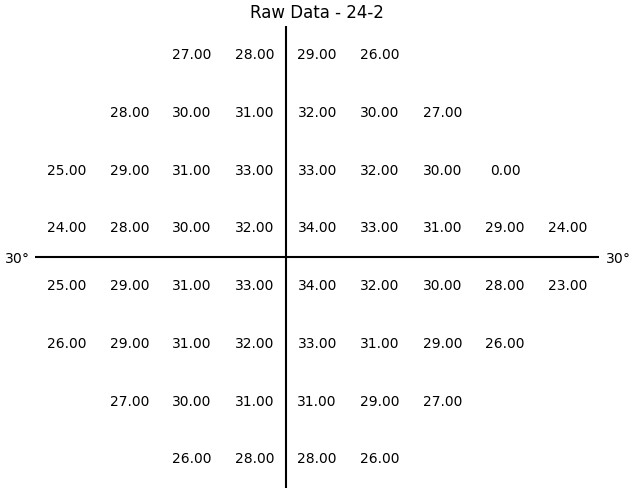
Sources: Waisberg et al. (2023); Ong et al. (2022). RAPD = relative afferent pupillary defect; ICP = intracranial pressure.

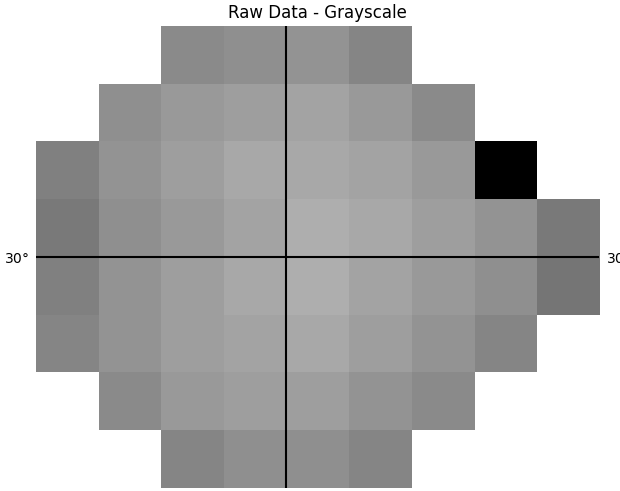
What the output looks like

Pupil diameter over time, both eyes (RAPD test)



Visual field map (24-2 pattern)





Example output from a healthy participant in a terrestrial clinic deployment. Illustrative only; not flight data.

03

Going to space

The headset is currently in space on the ISS!

Initial Design Contstraints

1

Self-administered

Astronauts can run every test alone, if desired

2

Scarce crew time

Every minute of the exam has to count

3

No hands-on fixes

Difficult to adjust the device once it is aboard

4

Same exam on ground and orbit

Identical protocol so in-flight results compare with terrestrial baseline data

Working with NASA, we collected baseline data from astronauts, and the platform is now in space.

In microgravity, the headset's own tracking failed

WHAT HAPPENED

- **Tumbling menu**
- **Nothing usable:** nothing could be seen or selected
- **No controllers:** the hand controllers did not track
- **No fix after launch:** settings such as travel mode or kiosk mode could not be enabled once aboard



Two things kept the tests running

FIX 1 · REMOTE CONTROL

Android Debug Bridge (ADB)

Apps were started and stopped from the command line, bypassing the headset's menus and controllers.

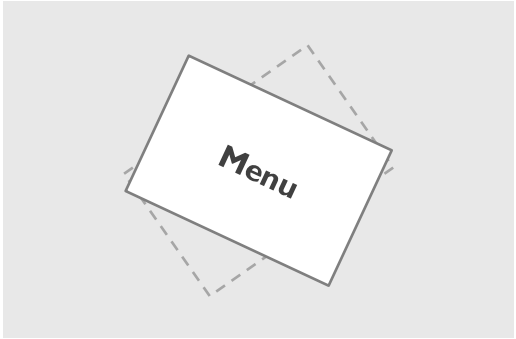


Headset menus and controllers bypassed

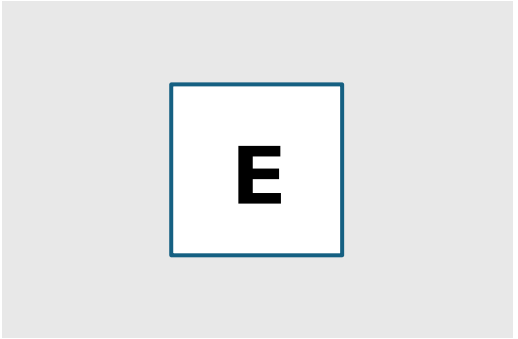
FIX 2 · HEAD-LOCKED TESTS

Headset-anchored stimuli

Every test display was attached to the headset in Unity, so it stayed in front of the eyes even when world tracking failed.



World-anchored: tumbles
System menu



Head-anchored: stays put
Our tests

04

Lessons learned

What we would tell anyone building VR tools for altered gravity

Four lessons for building VR tools for micro gravity

LESSON 1
Reliable sensing

THE LESSON

Don't assume gravity-dependent tracking will work

WHAT WE SAW

The headset's own tracking failed in microgravity

LESSON 2
User interaction

THE LESSON

Plan for no controller pointing: use gaze, the headset button, or buttons only

WHAT WE SAW

The controllers did not track in orbit

LESSON 3
Protocol robustness

THE LESSON

Attach stimuli to the headset; launch straight into the test

WHAT WE SAW

Head-locked tests kept working while the system menu tumbled

LESSON 4
Hardware–software integration

THE LESSON

Lock device settings (kiosk/travel mode) before launch and build in a remote control path such as ADB

WHAT WE SAW

Settings could not be changed once aboard

Takeaways

01

Moon and Mars crews will need eye exams they can run on board

NASA rates these eye changes a top risk for a crewed Mars mission

02

A single headset can deliver a standardized, multi-test eye exam targeted at SANS

One headset runs five tests with controlled light and both eyes tracked

03

Microgravity disrupts consumer headsets in ways design can work around

Head-locked stimuli and remote control kept the exam running after tracking failed

Next steps: in-flight data collection, head-down-tilt bed rest testing and test–retest reliability

Sources: Waisberg et al. (2023); Sarker et al. (A) (2023), Transl. Vis. Sci. Technol.; Patel et al. (2020), NPJ Microgravity

IVRHA SYMPOSIUM

Thank you!

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Joshua Ong, MD

LET'S COLLABORATE

Interested in this work? Let's talk.

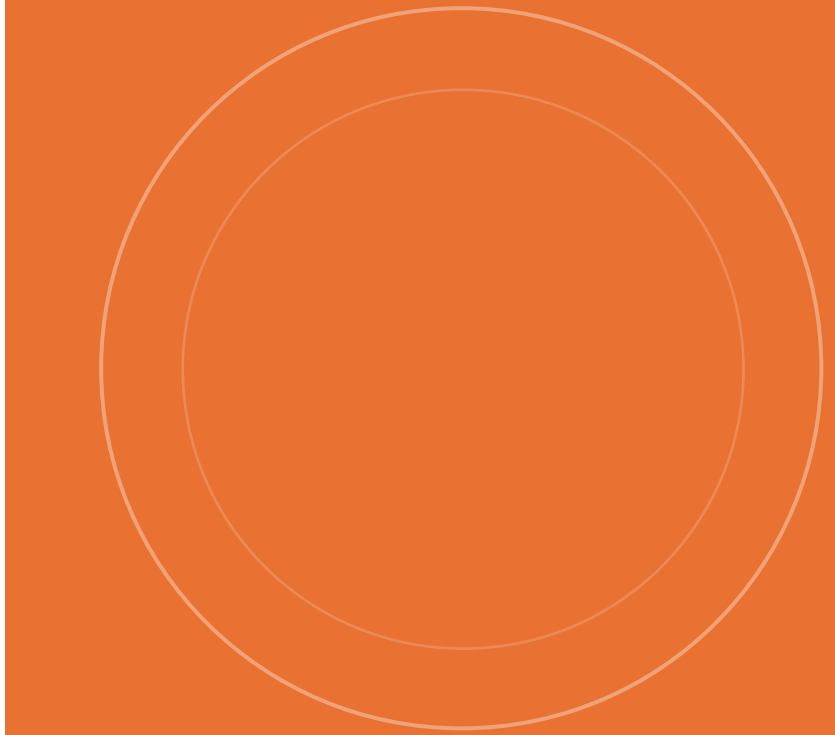
Working on VR, space health or eye care? I'd love to chat.

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DISCUSSION

Questions?

Joseph Tran
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Acuity: tracking the farsighted shift

THE CLINICAL TEST

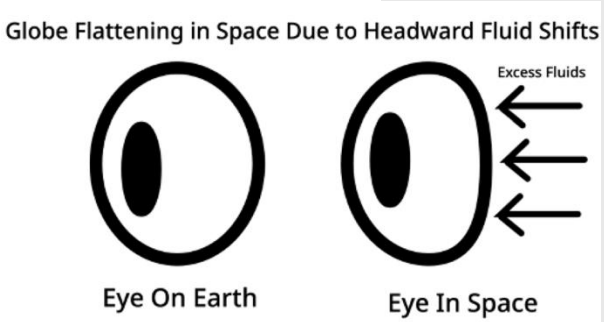
Visual acuity: tumbling E

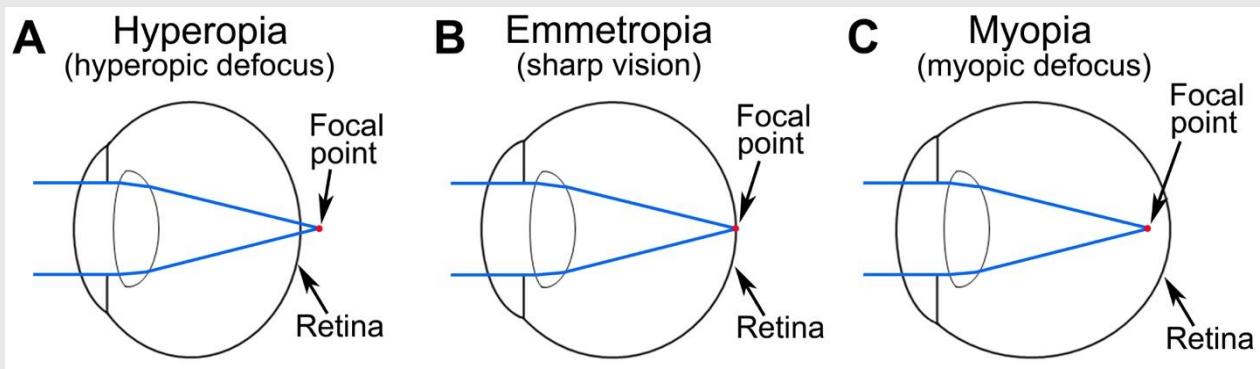
The letter E appears facing one of four directions and gets smaller each round.

The smallest E the user can still orient sets the acuity score.

THE EVIDENCE

- **Globe flattening shortens the eye** and causes a hyperopic (farsighted) shift[1]
- **VR acuity is proposed** to track that shift over a mission[1]
- **Caveat: headsets have a fixed focal distance,** so compare each person to their own baseline rather than using the test for diagnosis





1. Ong et al. (2022).

Contrast: the spaceflight contrast sensitivity hypothesis

THE CLINICAL TEST

Contrast sensitivity: sinusoidal gratings

Striped patterns appear at several spatial frequencies, from coarse to fine stripes.

Contrast drops until the user can no longer see the stripes.

THE EVIDENCE

- **Prolonged disc edema may damage retinal ganglion cells** and lower contrast sensitivity[1]
- **In IIH, peripheral contrast is hit harder** than central contrast[2]
- **When papilledema improves,** contrast sensitivity improves while Snellen acuity does not[3]
- **Contrast is also vital** for mission tasks

1. Waisberg et al. (2023). 2. Rehman et al. (2022). 3. Wall (1990).

Color: an exploratory measure of optic nerve function

THE CLINICAL TEST

Color vision

Ishihara plates: dot patterns that screen for red-green deficits.

Cambridge Color Test: measures the smallest color difference the user can detect (chromatic discrimination threshold).

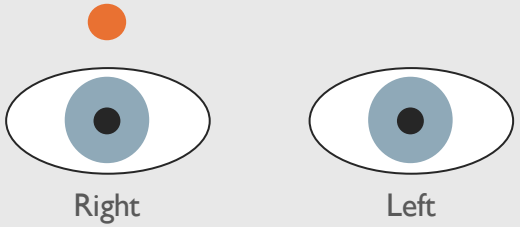
THE EVIDENCE

- **The parvocellular pathway carries color** and may be affected by ganglion cell damage[1]
- **No astronaut color findings have been reported,** and in IIH color vision did not change as papilledema resolved[2]
- **The Cambridge Color Test gives a quantitative threshold** for tracking small changes

1. Waisberg et al. (2023). 2. Wall (1990).

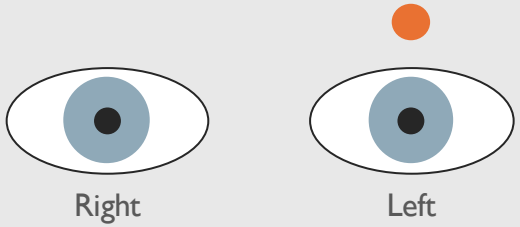
RAPD and pupils: comparing the optic nerves

1 Light on right eye



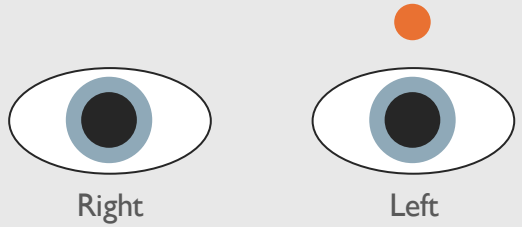
Both pupils constrict together

2a Swing to left: normal



Both pupils stay small

2b Swing to left: RAPD



Both pupils widen: the left optic nerve sends a weaker signal

THE EVIDENCE

- **SANS disc swelling is more often uneven** than in IIH, so comparing the two optic nerves matters[1]
- **Pupil reactivity tracks ICP in clinical patients:** high-ICP patients constrict and dilate more slowly[2]
- **VR RAPD measurement** has published test–retest reliability data[3]

1. Waisberg et al. (2023). 2. Clinical studies cited in Sarker et al. (2023), Life Sci. Space Res. 3. Sarker et al. (2023), Transl. Vis. Sci. Technol.
Test: Broadway (2012); VR version builds on Sarker et al. (A) (2022). RAPD = relative afferent pupillary defect; ICP = intracranial pressure. Orange dot = lit eye.

Visual fields: beyond the Amsler grid

THE CLINICAL TEST

Visual field perimetry: 24-2 pattern

The user fixates a central target while small lights appear across the central 24° of vision.

Missed points map where sensitivity is reduced, including the blind spot.

THE EVIDENCE

- **Perimetry is the standard follow-up** for papilledema on Earth[1]
- **Returning astronauts haven't shown reproducible field defects,** but standard testing may miss subtle change[2]
- **The Amsler grid used on the station** covers only the central field[3]

1. Wall (1990). 2. Huang et al. (2019). 3. Waisberg et al. (2023).

Is ICP actually elevated in flight? Still open.

WHAT THE MEASUREMENTS SHOW

- **Post-flight lumbar punctures:** normal to slightly elevated pressures, sometimes measured long after flight[1]
- **Parabolic flights:** no ICP rise above the supine level[1]
- **But parabolic flights last minutes,** not months

WHAT IT MEANS FOR OUR PLATFORM

- **Elevated ICP in flight** is a leading hypothesis, not a measured fact
- **Pupillometry is a candidate** non-invasive index of ICP, not a proven one
- **Our RAPD test records pupil reactivity,** so it can help test the hypothesis over time

1. Waisberg et al. (2023) ICP = intracranial pressure.

ICP has never been measured on the ISS because the reliable methods are invasive

INVASIVE: NOT POSSIBLE IN FLIGHT

Lumbar puncture

Needle into the lower spine. Astronauts only had it after landing, with no pre-flight baseline

Ventricular reservoir (Ommaya)

Implanted port into the brain. Used in 8 patients during brief parabolic flights

Epidural monitor

Implanted sensor. The only in-orbit recording is one rhesus monkey, 1992

NON-INVASIVE CANDIDATES

- CT imaging
- Tympanic membrane displacement
- Near-infrared spectroscopy
- Blood pressure + transcranial Doppler
- Fundoscopy (optic disc exam)
- **Pupillometry** what our RAPD test records

Sources: Sarker et al. (A) (2023), Table 1; Nguyen et al. (2025), Section 6.1.

Backup: how SANS is studied and countered

HOW RESEARCHERS STUDY SANS

- **Head-down tilt bed rest:** 6° tilt on Earth reproduces disc swelling, folds and globe flattening
- **Dry immersion:** floating in water mimics the headward fluid shift
- **Parabolic flight:** 20–25 s of weightlessness; direct ICP readings
- **In-flight OCT:** tracks optic disc and retinal swelling across the mission
- **Brain MRI:** CSF, venous and brain-position changes after flight
- **Genetics:** B-vitamin (one-carbon) gene variants raise SANS risk

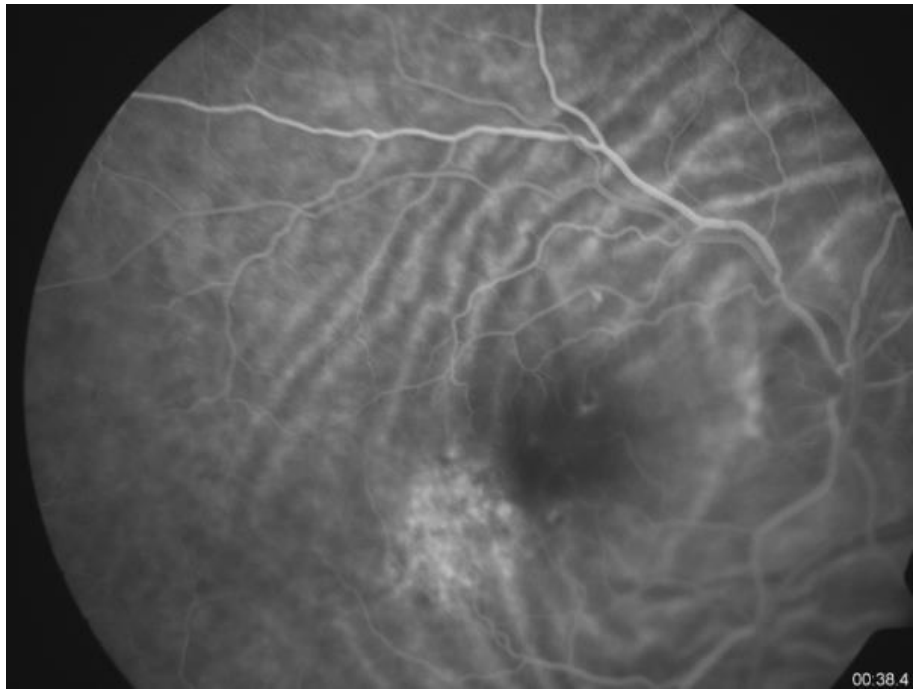
COUNTERMEASURES BEING TESTED

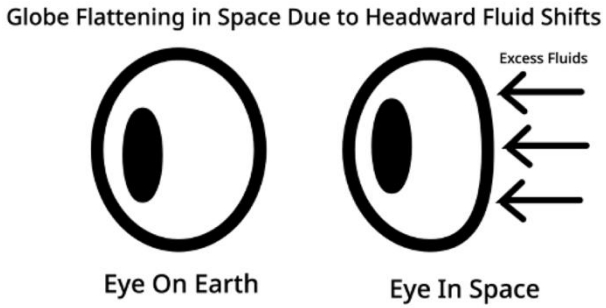
- **LBNP:** suction pulls fluid back to the legs
- **Thigh cuffs:** trap blood in the legs; used since the 1960s
- **Impedance threshold device:** breathing resistor; lowered ICP
- **Pressurized goggles:** rebalance pressure at the optic nerve
- **Centrifugation:** short bursts of artificial gravity
- **B-vitamin supplements:** for genetically at-risk astronauts
- **Space anticipation glasses:** the only proven fix (for vision shift)

Source: Nguyen et al. (2025), Progress in Retinal and Eye Research, sections 5–7.

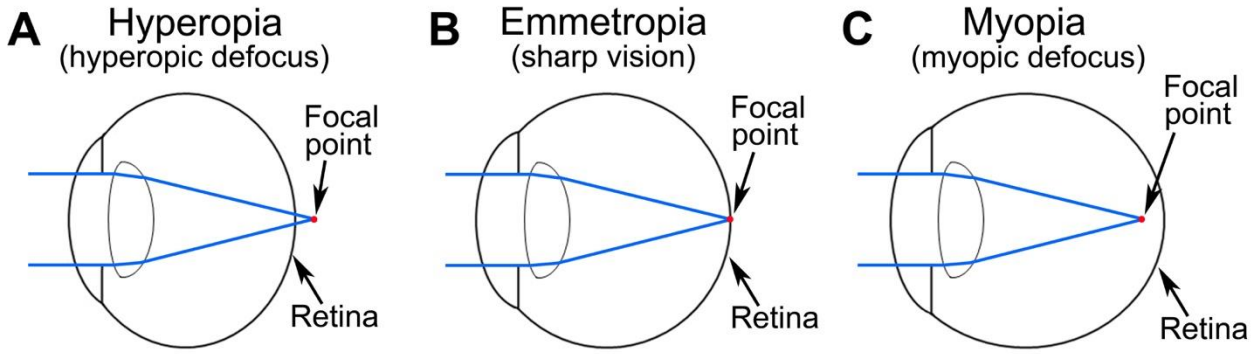
SANS Symptoms

Chorioretinal folds





More farsighted



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